

Natural Sources

Life on this planet earth depends upon large number of things and services provided by the nature, which are known as Natural resources.

Thus water, air, soil, minerals, coal, forests, crops and wild life are all examples of natural resources

- The natural resources are of two kinds :
- Renewable resources which are inexhaustive and can be regenerated within a given span of time e.g. forests, wildlife, wind energy, biomass energy, tidal energy, hydro power etc.
- Solar energy is also a renewable form of energy as it is an inexhaustible source of energy.

- Even our renewable resources can become non-renewable if we exploit them to such extent that their rate of consumption exceeds their rate of regeneration. and recycling.
- There are four major types of nonrenewable resources: oil, natural gas, coal, and nuclear energy. You may add even mineral resources ?
- Answer ?

Sustainable use of Resources

- It is very important to protect and conserve our natural resources and use them in a judicious manner so that we do not exhaust them.
- This is very important for the sustainable development.

- **Major natural resources:**
- (i) Forest resources
- (ii) Water resources
- (iii) Mineral resources
- (iv) **Food resources – Not to be covered as per syllabus of this course.**
- (v) Energy resources
- (vi) Land resources.

FOREST RESOURCES

- About 1/3rd of the worlds land area is forested which includes closed as well as open forests.
- Former USSR accounts for about a 5th of the worlds forests (20 %) , Brazil for about a 7th (around 15 %), and Canada and USA each for 6-7%. India just 2 %
- Almost everywhere the cover of the natural forests has declined over the years.

Forest Area by Country

- # Country Square meters per capita
- 1 Russia 55,992
- 2 Brazil 23,652
- 3 Canada 94,461
- 4 United States 9,556
- 5. India 600

- The greatest loss occurred in tropical Asia where one third of the forest resources have been destroyed.

USES OF FORESTS

Commercial uses:

Firewood, pulpwood, food items, gum, resins, non-edible oils, rubber, fibers, lac, bamboo canes, fodder, medicine, drugs and many more items, the total worth of which is estimated to be more than \$ 300 billion per year

Forest



Ecological uses

- **Production of oxygen:**
- **Reducing global warming:**
- **Wild life habitat:**
- **Regulation of hydrological cycle:**
- **Soil Conservation:**
- **Pollution moderators:**

Maintaining Tiger Count in India : Around 3000



Restoring ecosystem- **Role of Royal Hunters ?**

- Introduced 8 + 12 = 20 cheetahs brought from Namibia in Kuno Park, MP, these were declared extinct in 1952 in India .53 now 2026 .Anout 33 born in India.



- **Total Recorded Forest Area in the country is 765,210 sq. km, which is 23.28% of total geographical area of the country.**
- **Recorded Forests can be broadly classified into three categories** **Reserved Forests** (416,516 sq. **Protected Forests** (223,309 sq. km) (**declared by State Govts.**) **and** **Unclassed Forests** (125,385 sq. km).

- **Area-wise Madhya Pradesh has the largest forest cover in the country followed by Arunachal Pradesh, Chhattisgarh, Odisha and Maharashtra.**

The Larger Forests in India

- **Abujmarh.** Spread over about 3,870 Km, the Abujmarh is a forest which is situated in Chhattisgarh. ...
- **Anekal Reserved Forest.** Not much is known about this forest. ...
- **Baikunthapur Forests.** ...West Bengal
- **Bhavnagar, Amreli Forest (Gir)** ...
- **Bhitarkanika Mangroves.** ...
- **Dvaita Forest.** ...
- **Jakanari Reserved Forest.**

Different Forests in India (List) / May be deleted

- **Name** **Location** **Area**
- **Gir National Park** **Talala taluk, Gir Somnath district, Gujarat** **1412 km²**
- **Jakanari reserve forest** **Coimbatore**
- **Jim Corbett National Park** **Nainital district and Pauri Garhwal district, Uttarakhand** **520.8 km²**
- **Kanha National Park** **Madhya Pradesh** **650 km²**

May be deleted

- **Abujmarh forest** **Chhattisgarh**
- **Annekal Reserved Forest** **Western Ghats - Kerala, Tamil Nadu, Karnataka, Goa, Maharashtra and Gujarat.**
- **Baikunthapur Forest** **Dooars, West Bengal**
This is a terai forest
- **Bandipur National Park** **Karnataka 874 km²**
- **Bhagwan Mahaveer Sanctuary and Mollem National Park** **Sanguem taluk, Goa 650 km²**

TERAI

- The Terai is a belt of marshy grasslands, savannas (Scattered trees) and forests located south of the outer foothills of the Himalaya, the Shivalik Hills, and north of the Indo-Gangetic Plain of the Ganga, Brahmaputra and their tributaries.
- The Terai belongs to the Terai-Duar savanna and grasslands eco-region.

Family visits to enjoy and understand the Biodiversity – choose - interest

- **Bhitarkanika Mangroves**
- **Bhitarkanika lakeride sunset, Odisha 650 km²**
- **Bondla Wildlife Sanctuary Ponda taluk, Goa 8 km² -**

Provides sanctuary to leopards who have been injured in human-wildlife conflict.

- **Cotigao Wildlife Sanctuary Canacona taluk, Goa -**

It is known for its dense forest of tall trees (30 meters).

Record for Conservation and sustainability.

- **Gir National Park** - Talala taluk, Gir Somnath district, Gujarat 1412 km²
- **Jakanari reserve forest,** Coimbatore
- **Jim Corbett National Park, Nainital district and Pauri Garhwal district, Uttarakhand** 520.8 km²
- Some have been repeated here as a part of Tabulation under Different Forests of India).
Please note. People from far and wide visit these .

- **Kanha National Park** - **Tiger Kanha National Park.**
 - **Madhya Pradesh** 650 km² The present-day Kanha area is divided into two sanctuaries, Hallon and Banjar, of 250 and 300 km² respectively.

Kaziranga National Park is a protected wildlife reserve and UNESCO World Heritage Site located in the Golaghat, Nagaon, and Sonitpur districts of Assam, India. Location: Assam, along the Brahmaputra River floodplain.

Area: Approximately 430 square kilometers

Established: 1905 as a reserve forest; declared a national park in 1974

Keibul Lamjao National Park - Bishnupur district, **Manipur**
40 km² - The national park – **The only Floating park in the World.**
11 km² - **World's second biggest mangrove forest**

May be deleted

- Decomposed plant materials locally called phumdis.
- **Kukrail Reserve Forest** Lucknow, Uttar Pradesh
- **Mhadei Wildlife Sanctuary** Sattari taluk, Goa
208.5 km² Bengal tigers can be found here.
- **Molai forest** Majuli island on the Brahmaputra River 5.5 km²- Assam
- **Nagarhole National Park**

Rows

- **Elephant herd** at Nagarahole wildlife sanctuary.

Kodagu district and Mysore district, **Karnataka**
- 642 km² **One of India's premier Tiger Reserves**

- **Nallamala Hills**, Hill on Top of Nallamalla Hills temple.jpg **Eastern Ghats, Andhra Pradesh** (South of River Krishna).

May be deleted

- **Namdapha National Park** **Arunachal Pradesh** **1985**
km² **Fourth largest national park in India.**
- **Nanmangalam forest** **Chennai, Tamil Nadu** **24**
km² **The reserve forest area is 3.2 km²**
- **Netravali Wildlife Sanctuary** **Goa** **211 km²**
- **New Amarambalam Reserved Forest** **Nilambur,**
Malappuram district, Kerala
- **Pichavaram Mangrove Forest**
- **Pichavaram, Cuddalore district, Tamil Nadu.**



OVER EXPLOITATION OF FORESTS

- **The international timber trade alone is worth over US \$ 40 billion per year.**
- **Excessive use of fuelwood and charcoal, expansion of urban, agricultural and industrial areas and overgrazing have together led to over-exploitation of our forests leading to their rapid degradation.**

DEFORESTATION- Cause of Great Concern

- **The total forest area of the world in 1900 was estimated to be 7,000 million hectares which was reduced to 2890 million ha in 1975 and fell down to just 2,300 million ha by 2000.**
- **At the present rate it is estimated that in the next 60 years we would lose more than 90 percent of our tropical forests - Tree Transplantation, Plant 10 saplings for felling one tree.**

- **The forested area in India seems to have stabilized since 1982.**
- **As per our National Forest Policy, as we are still having only 19.27 % of our land area (63.38m ha) covered by forests based on satellite data (MoEF, 1998).**
- **Presently around 22%.**

Major Causes of Deforestation

- (i) **Shifting cultivation**: It is a farming in which the farmer cultivates on a plot of land temporarily, and when he finds the land infertile to grow due to soil exhaustion burns and shifts.
- In India, we have this practice in **NorthEast** and to some extent in **Andhra Pradesh, Bihar** and **M.P** which contribute to nearly half of the forest clearing annually.

(ii) Fuel requirements: Growing demand for fuelwood by increasing population in India.

(iii) Raw materials for industrial use: Wood for making boxes, furniture, railway-sleepers, plywood, match-boxes, pulp for paper industry etc. have exerted tremendous pressure on forests.

- **(iv) Development projects:** Various Development projects like hydroelectric projects, big dams, road construction, mining etc.
- **(v) Growing food needs:** To meet the demands of rapidly growing population, agricultural lands and settlements are created permanently by clearing forests.

- **Overgrazing:** The forest cleared lands for excessive fuelwood are turned into the grazing lands.
- **Overgrazing by the cattle leads to further degradation of these lands.**

Major Consequences of Deforestation

- (1) This threatens the existence of many wild life species.
 - (ii) Biodiversity is lost and along with that genetic diversity is eroded – Cheetahs became extinct in India, several birds become extinct, tree varieties are lost.
 - (iii) Hydrological cycle gets affected, thereby influencing rainfall- Transpiration.
 - (iv) Problems of soil erosion and loss of soil fertility increase.
 - (v) In hilly areas it often leads to landslides

Major Activities in Forests

- **Timber Extraction:** Teak and Mahogany wood and Sandal wood leads to removal of other trees. Construction of approach roads.
- **Mining:** Mining operations for extracting minerals and fossil fuels like coal often involves vast forest areas.
- Large scale **deforestation** has been reported in **Mussorie and Dehradun valley** due to **indiscriminate mining** of various minerals.

- **Indiscriminate mining** in forests of Goa
- Coal mining in Jharia, Raniganj etc. – **Jharia coal Fires?**
- Mining of magnesite and soap- stones (magnesium rich mineral talc.) . Kosi valley, Almora.
- Mining of radioactive minerals in Kerala, Tamil Nadu and Karnataka.

Western Ghats are also facing the same threat due to mining projects for **excavation of** copper, chromite, bauxite and magnetite.

Video on What are Natural Resources

- https://www.google.com/search?q=video+on+natural+resources&sca_esv=ac3607b23cf7e9a0&rlz=1C1YTUH_enIN1045IN1045&biw=1507&bih=655&sxsrf=APpeQns_Q-8csZthTUTEHiYLDLDju-mfPA%3A1788501765720&ei=BV-aasuxK_eb4-EPulbg4AI&ved=2ahUKEwiLyLOjoNSWAxX3zTgGHTgDGCwQ4dUDegQIBhAM&uact=

DAMS AND THEIR EFFECTS ON FORESTS AND PEOPLE

- **India has more than 1550 large dams,**
- **The highest one is Tehri dam, on river Bhagirathi in Uttaranchal and the largest in terms of capacity is Bhakra Nangal dam on river Satluj in H.P.**

- The crusade against the ecological damage and deforestation caused due to Tehri dam was led by

Sh. Sunder Lal Bahuguna, the leader of **Chipko movement**. The cause of Sardar Sarovar Dam related issues has been taken up by the environmental activists **Medha Patkar**, joined by **Arundhati Ray** and **Baba Amte**.

- **For building big dams, large scale devastation of forests takes place which breaks the natural ecological balance of the region.**
- **Floods, droughts and landslides become more prevalent in such areas.**

WATER RESOURCES

- India accounts for 18% of the world population and about 4% of the world's water resources..
- About 97% of the earth's surface is covered by **water** and most of the animals and plants have 60-65% water in their body.
- **Unique features of water**
 - 1. Liquid over a wide range of temperature..
 - 2. Due to **highest specific heat**, it warms up and cools down **very slowly** without causing **shocks of temperature jerks** to the aquatic life.

3. Due to high latent heat of vaporization it produces a cooling effect as it evaporates.
4. It is an excellent solvent for several nutrients including oxygen.
5. Due to high surface tension and cohesion it can easily rise through the trunk even in the tallest of the trees like Sequoia .
6. Some oxygen and CO₂ can dissolve in water and remain as the that helps in the survival of aquatic animals.
dissolved oxygen (DO)
7. Though concentration of CO₂ in water is poor but the photosynthetic efficiency of algae is more than that of terrestrial plants and this is due to CCM i.e., Carbon concentrating mechanism in the algae that takes place in chloroplasts with the final concentration of CO₂ occurring inside a specialized sub-compartment called the pyrenoid.

Sequoia Trees (About 84 m high)



- 8. It has an anomalous expansion behaviour (4 Deg. C TO 0 Deg. C i.e. as it freezes, it expands instead of contracting and thus becomes lighter and in extreme cold, the lakes freeze only on the surface.
- Aquatic animals can survive.
- 9. Electrolyses to give hydrogen.
- 10. Essential for all life on the earth.

Being lighter the ice keeps floating, whereas the bottom waters remain at a higher temperature and therefore, can sustain aquatic organisms even in extreme cold.

OTEC – Ocean Thermal Energy Conversion – Power Generation in the sea and ocean.

- The water we use keeps on cycling endlessly through the environment, which we call as **Hydrological Cycle**. We have enormous resources of water on the earth **amounting to about 1404 million Km³**.
- Every year about 1.4 inch thick layer of water evaporates from the oceans, **more than 90%** of which returns to the oceans through the **hydrological cycle**.

- **Six countries—Brazil, Russia, Canada, Indonesia, China, and Colombia—account for half of Earth's freshwater supply.**
- **Water poor countries : Bahrain, Jordan, Kuwait, Libyan Arab Jamahiriya, Maldives, Malta, Qatar, Saudi Arabia, United Arab Emirates , Yemen.**

Major River Systems in India

- **1. Indus River System.**
- **2 Brahmaputra River System.**
- **3 Ganga River System.**
- **4 Yamuna River System.**
- **5 Narmada River System.**
- **6 Tapti River System.**
- **7 Godavari River System.**
- **8 Krishna River System.**
- **The Brahmaputra River is the largest river in India by water volume, but the Ganges River is the longest by length.**

Rivers of India



WATER USE AND OVER-EXPLOITATION

- Water use by humans is of three types:
- 1. Water withdrawal: Taking water from groundwater or surface water resource
- 2. water consumption: The water which is taken up but not returned for reuse – evaporated or retained in plants.
- 3. Hydrogen generation.
- With increasing human population and rapid development, the world water withdrawal demands have increased many folds.
- A large proportion of the water withdrawn is polluted due to anthropogenic activities.

Water footprint

- On a global average 70 percent of the water withdrawn is used for agriculture.
- In India, we use 93% of water in agricultural sector while in a country like Kuwait, which is water-poor, only 4% is used for watering the crops.
- About 25% of water on global average is used in industry, which again varies from a high of 70% in European countries to as low as 5% in less developed countries.

Water export through crops export

- India is the world's 2nd largest producer and eighth largest wheat exporter and accounted for 4.1 percent of total global wheat exports in 2020-21. Good quality wheat is produced & there is demand.
- India is the world's second-largest rice producer, after China, accounting for 40% of the global rice trade. It exports the grain to over 150 countries – **Water is thus exported through crops (Virtual Water Export).**
- **India has witnessed a continued decline in its per capita water availability — by 60 per cent over the last 50 years because of increasing water use by the irrigation sector.**
- **Thus, crops, meat, leather, chemicals and industrial goods are imported to ensure a positive water balance.**

- **In USA, an average family of 4 consumes more than 1000 M3 of water per year, which is many times more than that in most developing countries.**
- **Domestic water consumption in India is generally estimated at 135 liters per capita per day (lpcd) for urban areas and 75 lpcd for rural area.**

Water: A Precious Natural Resource

- Out of the total water reserves of the world, about 97% is salty water (marine) and only 3% is fresh water.
- Most of it is locked up in polar ice caps and just 0.003% is readily available to us in the form of groundwater and surface water
- Overuse of groundwater for drinking, irrigation and domestic purposes has resulted in rapid depletion of groundwater in various regions leading to lowering of water table and drying of wells. Water poor states in India ?

As, Fluorides, Iron, Nitrates etc. in Groundwater

- **Pollution of many of the groundwater aquifers has made many of these wells unfit.**
- **Unfortunately, growth of population, has been responsible for pollution of the rivers for consumption.**
- **Out of 8 - 9 billion people, some 2.2 billion people around the world do not have safely-managed drinking water, while 4.2 billion go without safe sanitation services and 3 billion lack basic handwashing facilities, according to a new report from the United Nations Children's Fund (UNICEF) and the World Health Organization (WHO).**

In India people walk 5-20 kms just to fetch water in some rural areas. (UP to 78 meters deep even?)

- It was estimated that by 2024, two-thirds of the world population would be suffering from acute water shortage due to increase in population and industrialization- 78 METERS



- **The 2025/2026 Reality: The United Nations and UNESCO confirm that between 2 and 3 billion people face seasonal shortages, but when factoring in pollution and infrastructure gaps, the number under water stress hits the 4 billion mark.**

Groundwater – More water

- **About 9.86% of the total fresh water resources is in the form of groundwater and it is about 35-50 times that of surface water supplies.**
- **These are also being polluted.**
- **A layer of sediment or rock that is highly permeable and contains water is called an aquifer.**

- **Unconfined aquifers** - which are **overlaid by permeable earth materials** and they are recharged by water seeping down from above in the form of **rainfall and snow melt**.
- **Confined aquifers** - which are **sandwiched between two impermeable layers of rock or sediments** and are recharged only in those areas where **the aquifer intersects the land surface**.

Effects of Groundwater Usage

- **(a). Subsidence:** When groundwater withdrawal is more than its recharge rate, the sediments in the aquifer get compacted, a phenomenon known as ground subsidence. There is sinking of earth and damage to buildings, pipelines etc.
- (b). Lowering of water table:** Mining of groundwater is done extensively in **arid and semi-arid regions** for irrigating crop fields.

- The decline of groundwater levels in India by 5-10 m is usually explained as a result of the increased number of tube wells and a decline in average annual rainfall.
- Haryana, Punjab ?
- Scientists using data from NASA's Gravity Recovery and Climate Experiment (GRACE) have found that the groundwater beneath Northern India has been receding by as much as one foot per year over the past decade.

- **Water logging** : When excessive irrigation is done with brackish water it raises the water table gradually leading to water-logging and **salinity** problems.
- **Surface Water** -

This is the water coming through precipitation (rainfall, snow) **when this does not percolate down into the ground. Or this does not return to the atmosphere as evaporation or**

or transpiration loss, assumes the form of streams, lakes, ponds, wetlands or artificial reservoirs is known as surface water.

- The surface water is largely used for irrigation, industrial use, public water supply, navigation etc.

Water Rich vs. Water Poor countries

- The top 10-11 water rich countries are Iceland, Surinam, Guyana, Papua New Guinea, Gabon, Solomon Islands, Canada, Norway, Panama, Columbia and of course Brazil lying in the far north and have low evaporation losses.

- The water poor countries include Kuwait, Egypt, United Arab Emirates, Malta, Jordan, Saudi Arabia, Singapore, Maldives, Israel and Oman, lying in the desert belt at about 15° to 25° Latitude and some of them like Malta and Singapore are densely populated areas resulting in low per capita water.

Ottawa River - The river is 1,271 km (790 mi) long; it drains an area of 146,300 km² (56,500 sq mi), 65 per cent in Quebec and the rest in Ontario,

FLOODS

- In some countries like India and Bangladesh rainfall does not occur throughout the year, rather, **90% of it is concentrated into a few months (June-September).**
– **Monsoon.**
- Deforestation, overgrazing, mining, rapid industrialization, global warming etc. have also **contributed largely to a sharp rise in floods.**
- Cloud bursts also result in floods – **Kedarnath – June 2013, 41 persons died., 4550 villagers affected.**
Present situation this year? Cloud bursts ?
- Excess eainfall in 2025, 2026 ?

DROUGHTS

- There are **about 80 countries in the world,** lying in the **arid and semiarid regions** that experience frequent spells of droughts, very often extending up to year long duration.
- **When annual rainfall is below normal and less than evaporation, drought conditions are created.**

Causes of Drought

- **Anthropogenic causes :**
- **Over grazing, deforestation, mining etc.** there is spreading of the **deserts** tending to convert more areas to drought affected areas.
- **India has experienced more and more desertification,** thereby increasing the vulnerability of larger parts of the country to droughts – **Around 70 % of area is dry land.**

- **Erroneous and intensive cropping pattern and increased exploitation of scarce water resources through well or canal irrigation to get high productivity has converted drought - prone areas into desertified ones, e.g., Maharashtra – 39 years and no recovery from Drought.**
- **Water contents of wheat and rice grains – around 14-20 %.**

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Remedial measures:

- Carefully selected mixed cropping help optimize production and minimize the risks of crop failures - For example, wheat + gram, wheat + mustard etc. crops are selected such that their nutrient requirements are different.
- Social Forestry and Wasteland development. Eucalyptus which is now known to lower the water table because of its very high transpiration rate and it has negative allelopathy .

CONFLICTS OVER WATER- May be deleted

- **Indispensability** of water and its unequal distribution has often led to inter-state or international disputes.
 - **Water conflict in the Middle East:**
 - **Three river basins, namely the Jordan, the Tigris-Euphrates and the Nile are the shared water resources for Middle East countries. Ethiopia controls the head waters of 80% of Nile flow. Egypt is being affected.**
- Dams and canals ?**

FAO – May be deleted

- India has pre-emptive right to construct barrages across all the rivers in Indian territory. However, the **Sindhu treaty requires that the three rivers allocated to Pakistan may be used for non-consumptive purposes by India i.e. without changing its flow and quality.** Recent reaction of India.

Under the IWT, (Indus Water Treaty) control over the **three eastern tributaries of the Indus River—Ravi, Sutlej, and Beas—is granted to India before they flow into Pakistan, and the three western tributaries—Indus, Jhelum, and Chenab—to Pakistan (FAO, 2011b).**

The Cauvery water dispute:

- : Out of India's 18 major rivers, 17 are shared between different states.
- In all these cases, **there are intense conflicts** over these resources which hardly seem to resolve.
- **Tamil Nadu, occupying the downstream region** of the river, wants water-use regulated in the upstream. Whereas, **the upstream state Karnataka** refuses to do so and claims its **primacy over the river as upstream user**.

The Satluj-Yamuna link (SYL) canal dispute:

- The issue of sharing the Ravi-Beas waters and SYL issue between Punjab and Haryana.
- Hathnikund barrage Problem for Delhi.
- Traditional Water Management System In India, even today, there are several villages where **water management is done** not by the Irrigation Department, **but by local managers.**

May be deleted

- In south India, a neerkatti manages the traditional tanks very efficiently based on his/her knowledge of the terrain, drainage and irrigation needs.
- In Maharashtra, the water managers are called havalgars or jaghyas who manage and resolve conflicts by overseeing the water channels from main canal to the distributory canals.

BIG DAMS- BENEFITS AND PROBLEMS

- **Benefits :**

- **1. Employment for tribals.**
 - **2. Checking floods and famines,**
 - **3. Generate electricity and reduce water and power shortage.**
- 4. Provide irrigation water to lower areas
provide drinking water in remote areas
and promote navigation, fishery etc.**

Environmental Problems

- The upstream problems include the following:
 - (i) Displacement of tribal people
 - (ii) Loss of forests, flora and fauna
 - (iii) Changes in fisheries and the spawning (laying eggs) grounds
 - (iv) Siltation and sedimentation of reservoirs
 - (v) Loss of non-forest land

- (vi) Stagnation and waterlogging near reservoir.
- (vii) Breeding of vectors and spread of vector-borne diseases – Dengue ?
- (viii) **Reservoir induced seismicity (RIS)** causing earthquakes.
- (ix) Growth of aquatic weeds.
- (x) Microclimatic changes.

- **(B) The downstream impacts include the following:**
- **(i) Water logging and salinity due to over irrigation.**
- **(ii) Micro-climatic changes**
- **(iii) Reduced water flow and silt deposition in river.**
- **(iv) Flash floods**
- **(v) Salt/soil water intrusion at river mouth**

- (vi) Loss of **land fertility** along the river since the **sediments carrying nutrients get deposited in the reservoir.**
- (vii) Outbreak of vector-borne diseases like malaria, dengue etc.
- **Thus, although dams are built to serve the society with multiple uses, but it has several serious side-effects. That is why now there is a shift towards construction of small dams or mini-hydel/ micro-hydel projects.**

MINERAL RESOURCES

- Minerals are naturally occurring, inorganic, crystalline solids having a definite chemical composition and characteristic physical properties – Silicates, carbonates, sulfates, halides, oxide etc.
- However, most of the rocks, we see everyday are just composed of a few common minerals like quartz, feldspar, biotite, dolomite, calcite, laterite etc.

- These minerals, in turn, are composed of some elements like oxygen, silicon, aluminium, iron,, calcium, magnesium etc

USES AND EXPLOITATION

- (i) Development of industrial plants and machinery.
- (ii) Generation of energy e.g. coal, lignite, uranium.
- (iii) Construction, housing, settlements.

- iv) Defence equipments - weapons, armaments.
- (v) Transportation means.
- (vi) Communication- telephone wires, cables, electronic devices.
- (vii) Medicinal system - particularly in Ayurvedic System.
- (viii) Formation of alloys for various purposes (e.g., phosphorite). – **Calcium phosphate**

- ix) Agriculture as fertilizers, seed dressings and fungicides (e.g. zineb containing zinc, Maneb-containing manganese etc. is a fungicide) – used in wheat storage and inhibiting rotting of fruit.
- (x) Jewellery e.g., Gold, silver, platinum, diamond.
- (xi) Permanent magnets of rare earth elements- Wind energy, Semiconductors etc.

Types of Minerals

- Based on their properties, minerals are basically of two types:
- (i) Non - metallic minerals e.g. graphite, diamond, quartz, feldspar etc.
- (ii) Metallic minerals e.g. Bauxite, laterite, haematite etc.
- Human history periods are named after them as Bronze (Cu – 88% and Sn-12 %) Age and Iron Age.

- The reserves of metals and the technical know-how to extract them have been the key elements in determining the economy and political power of nations.
- Li , other rare earth elements?
- Out of the various metals, the one used in maximum quantity is **Iron and steel** (740 million metric tons annually) followed by **manganese, copper, chromium, aluminium and Nickel.**

The 17 Rare Earth Elements:

Scandium (Sc) – Atomic number 21 **Yttrium (Y)** – Atomic number 39 **Lanthanum (La)** – Atomic number 57 **Cerium (Ce)** – Atomic number 58 **Praseodymium (Pr)** – Atomic number 59 **Neodymium (Nd)** – Atomic number 60 **Promethium (Pm)** – Atomic number 61 **Samarium (Sm)** – Atomic number 62 **Europium (Eu)** – Atomic number 63 **Gadolinium (Gd)** – Atomic number 64 **Terbium (Tb)** – Atomic number 65 **Dysprosium (Dy)** – Atomic number 66 **Holmium (Ho)** – Atomic number 67 **Erbium (Er)** – Atomic number 68 **Thulium (Tm)** – Atomic number 69 **Ytterbium (Yb)** – Atomic number 70 **Lutetium (Lu)** – Atomic number 71.

Critical Metals/Minerals

- Aluminum, used in almost all sectors of the economy
- Antimony, used in lead-acid batteries and flame retardants
- Arsenic, used in semiconductors
- Barite, used in oil and gas drilling and medical imaging
- Beryllium, used to manufacture metal alloys for aerospace and defense
- Bismuth, used in nontoxic metals, atomic research, and some medical applications
- Boron, used to harden steel and glass and in nuclear energy and solar energy etc. etc.
- List is long..... **You may look at it.**

Major reserves and important uses of some of the major metals

Metal	Major World Reserves	Major Uses
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Aluminium	Australia, Guinea, Jamaica	
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Uses : Packaging food items, transportation, utensils, electronics.

Chromium:

Kazakhstan, , South Africa, India

Uses : For making high strength steel alloys, In textile/tanning industries.

Cr (VI) ?

Copper:

U.S.A., Canada, Chile, Zambia

Uses :Electric and electronic goods,
building, construction, vessels etc.

Iron : Australia, Brazil, Russia, China

Uses : Heavy machinery, steel production,
transportation means.

Lead : Australia, China, Peru , Mexico, U.S.A.

- **Uses : Car batteries, radiation protection, paints, ammunition**
- **Manganese**
- **South Africa, Ukraine and Brazil**
- **Uses : For making high strength, heatresistant steel alloys..**

- **Platinum**

- South Africa, Canada, US, Zambia, Australia

- Uses : in automobiles, catalytic converters, electronics, medical uses.

- **Gold**

- Australia, Russia, South Africa, Indonesia

- Uses : Ornaments, medical use, electronic use, use in aerospace..

- **Silver :**
- Peru, Australia, Poland
- **Uses :** Photography, electronics , jewellery.
- **Cobalt :**
- DR Congo, Australia, Indonesia, Cuba, Philippines, Russia, Canada, Madagascar, China, USA.
- **Uses :** Aircraft engines, gas turbines, magnets, Super alloys , cemented carbides, catalysts.

- **Nickel** :
- Indonesia, Australia, Brazil, Russia, Cuba, , Philippines.
- Uses : Chemical industry, steel alloys, catalysts, batteries, coins.

Lithium : With 8 million tons, Chile has the world's largest known lithium reserves. This puts the South American country ahead of Australia (2.7 million tons), Argentina (2 million tons) and China (1 million tons).

Uses : Storage batteries for solar and wind power, rechargeable batteries for mobile phones, laptops, digital cameras and electric vehicles, catalysts (La and Ce) etc.

Rare Earth Metals

- Rare-earth ore deposits are found all over the world.
- The major ores are in **China, the United States, Australia, and Russia**, while other viable ore bodies are found in **Canada, India, South Africa, and southeast Asia**.
- Rare earth elements (REEs) and rare metals are key ingredients for glass, lights, magnets, batteries, and catalytic converters, and used in everything from cell phones to cars. For example, to make the magnet for one wind turbine, you need about 300 kilograms of neodymium.

Rare earth and energy are required for AI/ML in future

- Artificial intelligence (AI) depends heavily on hardware made with rare earths. These materials support everything from data storage to processors running complex learning systems. Dysprosium and terbium boost magnet strength in AI-powered robots.
- The other thing this would require is energy in future.
- Renewable energy also requires rare earth metals like

- **Rare earth elements are key to the development and utilization of renewable energy technology. Their properties allow the production of efficient solar panels, electric vehicles, wind turbines, energy storage devices, catalytic converters, and hydrogen production.**
- **<https://www.stanfordmaterials.com/blog/major-applications-of-rare-earth-elements-in-renewable-energy.html#:~:text=Rare%20earth%20elements%20are%20key,catalytic%20converters%2C%20and%20hydrogen%20production>.**

- **Wind Turbines are one of the most significant applications of REEs, particularly to generate wind energy. The production of wind turbines relies on REEs, including neodymium, praseodymium, dysprosium, and terbium, which are used to make high-strength permanent magnets.**
- **Rest you would learn though the Assignment 1.**

Major uses of some non-metallic minerals

- Silicate minerals : Sand and gravel for construction, bricks, paving
- Limestone : Used for concrete, building stone, used in agriculture for neutralizing acid soils, used in **cement** industry.
- Gypsum : Used in plaster wall-board, in agriculture, mixing in cement.

- Potash, phosphorite : Used as fertilizer
- Sulphur pyrites : Used in medicine, car battery, metallurgical industries

Australia, South Africa, the United States of America, Canada, etc. are having the major world reserves of most of the metallic minerals.

- The U.S. has abundant supplies of coal, copper, lead, iron, natural gas, timber, bauxite, and uranium.
- Gold
- Rich nations ?
- Why some countries are rich ?
- India' Golden sparrow- Padmanabhaswamy temple- Thiruvanthapuram, Kerala

- Due to huge Mineral and Energy resources, Scientific and Technological advancements (Maximum Nobel Prizes), the USA became the richest and the most powerful nation in the world in even less than 200 years.
- Japan too needs a mention here, as there are virtually no metal reserves, coal, oil and timber resources in Japan and it is totally dependent on other countries for its resources –
Japan is a technologically Advanced and is the 3rd largest Economy of the World.

- it has developed energy efficient technologies to upgrade these resources to high quality finished products to sustain its economy.
- Minerals are sometimes classified as Critical and Strategic for economic development.
- Critical minerals are essential for the economy of a nation e.g. iron, aluminium, copper, gold etc.
- Strategic minerals are those required for the defence of a country e.g. Manganese, cobalt, platinum, chromium etc.
- Lithium is important for Renewable Energy sources.
- Rare earth elements are essential for developing critical energy, transition technologies, and digitalization technologies, such as electric motors, wind turbines, and batteries [5]. High-tech devices, medical equipment, and military systems all rely heavily on rare earth elements (REEs).

Some Major Minerals of India- May be deleted

(a) Energy generating minerals -

Coal and lignite : West Bengal, Jharkhand, Orissa, M.P., A.P.

Uranium (Pitchblende or Uranite ore) : Jharkhand, Andhra Pradesh (Nellore, Nalgonda), Meghalaya, Rajasthan (Ajmer).

Other commercially used minerals

- **Aluminium (Bauxite ore)** : Jharkhand, West Bengal, Maharashtra, M.P., Tamil Nadu.

Iron (haematite and magnetite ore) : Jharkhand, Orissa, M.P., A.P., Tamil Nadu, Karnataka, Maharashtra and Goa.

- **Copper (Copper Pyrites)** : Rajasthan (Khetri), Bihar, Jharkhand, Karnataka, M.P., West Bengal, Andhra Pradesh and Uttaranchal.

Mineral Resources Rank of India

- India's major mineral resources include Coal (**4th largest reserves in the world**), Iron ore, Manganese ore and Bauxite (**7th largest**), Mica (**8th**), Chromite, Natural gas, Diamonds, Limestone and Thorium – Internet :
- Is India rich in Mineral resources?

[illegible]

**India's merchandise export in 2021-22
(April-January) was USD 335.44 billion
(B).**

- Refined Petroleum (\$25.3B),**
- Packaged Medicaments (\$17.8B),**
- Diamonds (\$16B),**
- Rice (\$8.21B),**
- Jewellery (\$7.57B),**
- Exporting mostly to United States (\$ 49.7 B),
China (\$18.5B), United Arab Emirates (\$ 18.1 B),
Hong Kong (\$ 9.18 B), and Germany (\$ 8.8 B).**

ENVIRONMENTAL IMPACTS OF MINERAL EXTRACTION AND USE

- **Indian Scenario :** India is the producer of 84 minerals the annual value of which is about Rs. 50,000 crore.
- **At least six major mines need a mention here which are known for causing severe problems**

- (i) Jaduguda Uranium Mine, Jharkhand exposing local people to radioactive hazards.
- (ii) Jharia coal mines, Jharkhand underground fire leading to land subsidence and forced displacement of people.
- (iii) Sukinda chromite mines, Orissa seeping of hexavalent chromium into river posing serious health hazard, Cr 6+ being highly toxic and carcinogenic.

- Kudremukh iron ore mine, Karnataka causing **river pollution** and threat to **biodiversity**.
- (v) East coast Bauxite mine, Orissa : Land encroachment and issue of rehabilitation unsettled.
- (vi) North-Eastern Coal Fields, Assam Very high sulphur contamination of groundwater

Impacts of mining:

Examples - Digging Sand , Gravel and rock for Gold.

Mining is done to extract minerals (or fossil fuels) from deep deposits in soil by using sub-surface mining or from shallow deposits by surface mining – Kolar Gold mines.

- **The former method is more destructive, dangerous and expensive including risks of occupational hazards and accidents.**

Surface mining can make use of any of the following three types

- :
- (a) **Open-pit mining** in which machines dig holes and remove the ores (e.g. copper, iron, gravel, limestone, sandstone, marble, granite).
- (b) **Dredging** in which chained buckets and draglines are used which scrap up the minerals from under-water mineral deposits.

- (c) **Strip mining** in which the ore is stripped off by using **bulldozers, power shovels and stripping wheels** (e.g. phosphate rocks).
- **The environmental damage caused by mining activities are as follows :**

- **(i) Devegetation and defacing of landscape:** The topsoil as well as the vegetation are removed from the mining area to get access to the deposit.
- **Large scale deforestation or devegetation leads to several ecological losses.**
- **(ii). Subsidence of land:** This is mainly associated with underground mining, e.g., tilting of buildings, cracks in houses, buckling of road, etc. .

- (iii). Groundwater contamination : **Mining disturbs the natural hydrological processes and also pollutes the groundwater.**
- Sulphur, usually present as an impurity in many ores is known to get converted into **sulphuric acid** through microbial action,
- Some heavy metals also get leached into the groundwater and contaminate it posing health hazards.

- (iv). Surface water pollution: The **acid mine drainage/ leachates** often contaminates the nearby streams and lakes.
- The acidic water is detrimental to many forms of aquatic life.
- (v) Air pollution : In order to separate and purify the metal from other impurities in the ore, **smelting is done** which emits enormous quantities of air pollutants damaging the **vegetation nearby and has serious environmental health impacts. .**

- (vi) **Occupational Health Hazards :** Most of the **miners** suffer from **various respiratory and skin diseases** due to constant exposure to the suspended particulate matter and toxic substances.
- (vii) **Noise and vibrations.**

Remedial measures: EIA, NABET, QCI

- Safety of mine workers should be the priority subject of industry.
- Adverse impacts of mining - Eco-friendly mining technology. should be adapted.
- The low-grade ores can be better utilized by using microbial-leaching technique using *Acidithiobacillus ferrooxidans* which may be more environment friendly.

- **Restoration of mined areas by re-vegetating them with appropriate plant species, stabilization of the mined lands, gradual restoration of flora, prevention of toxic drainage discharge and conforming to the standards of air emissions are essential for minimizing environmental impacts of mining.**

LAND RESOURCES

- **LAND AS A RESOURCE**
- Land is a finite and valuable resource upon which we depend for our food, fiber and fuel wood, the basic amenities of life.
- Soil, especially the top soil, is classified as a renewable resource because it is continuously regenerated by natural process though at a very slow rate.
- About 200-1000 years are needed for the formation of one inch or 2.5 cm soil, depending upon the climate and the soil type.

Country with most arable Land

- Arable land is defined by the United Nations Food and Agriculture Organization (FAO) as land currently used, or potentially capable of being used, to grow seasonal crops (Arable Acres)

In Acres

United States	389,767,633
India	385,641,557
Russia	300,594,679
China	295,220,748

Green Revolution in India- 1960

- Land use statistics by country
- Rank Country Arable land (%)
- 1 India 50.4
- 2 United States 16.8
- 3 Russia 7.3
- 4 China 11.3,
- Adoption of technology, such as the use of high yielding variety (HYV) seeds, mechanized farm tools, irrigation facilities, pesticides, and fertilizers.- Normam Borlaug – Dwarf wheat variety.
-

Explained before ?

• Rank	Country	Arable Hectares	Arable Hectares Acres
• 1	United States	157,736,800	389,767,633
• 2	India	156,067,000	385,641,557
• 3	Russia	121,649,000	300,594,679
• 4	China	119,474,200	295,220,748
• 5	Algeria	75,050,017	185,448,592
• 6	Brazil	55,762,000	137,787,902
• 7	Nigeria	34,000,000	84,014,000
• 8	Ukraine	32,924,000	81,355,204
• 9	Argentina	32,632,760	80,635,550
• 10	Australia	30,573,000	75,545,883

Total Areawise countries :

Algeria **185,448,592**

(<https://worldpopulationreview.com/country-rankings/arable-land-by-country>, Accessed on September 09, 2022).

Total Areawise countries :

1. Russia. Area: 17,125,000 square kilometres.
2. Canada. Area: 9,984,670 square kilometres.
3. China. Area: 9,572,900 square kilometres.
4. United States of America. Area: 9,525,067 square kilometres.
5. Brazil. Area: 8,515,767 square kilometres. ...
6. Australia. ...
7. India. ...
8. Argentina

LAND DEGRADATION

- With **increasing population growth** the demands for arable land for producing food, fibre and fuelwood is also increasing.
- **More population more land degradation due to overexploitation.**
- **The average annual erosion rate is 20-100 times more than the renewal rate.**

Degradation of Land

- **Soil erosion, water-logging, salinization and contamination of the soil with industrial wastes like fly-ash, press-mud or heavy metals all cause degradation of land.**
- **SOIL EROSION**
- **Soil erosion results in the loss of fertility of top soil layer.**
- **Two thirds of the seriously degraded lands lie in Asia and Africa.**

Type of Soil Erosion

- Soil erosion is of **two types**
- (i) **Normal erosion or geologic erosion** : caused by the **gradual removal of top soil** by natural processes. There is a balance between erosion and renewal.
- (ii). **Accelerated erosion**: This is mainly caused by **anthropogenic (man-made)** activities and the **rate of erosion is much faster than the rate of formation of soil.**

- **Overgrazing, deforestation and mining are some important activities causing accelerated erosion.**
- **AGENTS OF SOIL EROSION :**
- There are **two types of agents which cause soil erosion:**
- (i) Climatic agents: **Water and wind are the climatic agents** of soil erosion. Water affects soil erosion in the form of torrential rains, rapid flow of water along slopes, run-off, wave action and melting and movement of snow/ice.
- (ii). Biotic agents: Excessive grazing, mining and deforestation are the major biotic agents responsible for soil erosion.

The 7 methods of soil conserv

- These methods include
- (1). Installing biodegradable erosion control coir mats,(2).Practicing crop rotation, (3). implementing conservation tillage,(4).contour farming, (5). strip farming,(6). terrace farming, and (7). using diversion structures.
- Preventing soil erosion is essential for maintaining soil health, crop yield, and aquatic ecosystems.

SOIL CONSERVATION PRACTICES May be deleted

- **(i) Conservational till farming:**
or minimum tillage, is a broadly defined practice that includes no-till, strip till, ridge till, and mulch till systems. These techniques maintain plant residues (mulch) of previous crop on at least 30% of the soil surface after tillage activities.

•

Conservation tillage,



May be deleted

Inject seeds, fertilizers, herbicides and a little water in the slit, so that the seed germinates and the crop grows successfully without competition with weeds.

(ii) Contour farming: On gentle slopes, crops are grown in rows across, rather than up and down, a practice known as contour farming.

Each row planted horizontally along the slope of the land.

May be deleted

- Terracing :

It is used on still steeper slopes. These are converted into a series of broad terraces which run across the contour.

(iv) Strip cropping: Here strips of crops are alternated with strips of soil saving covercrops like grasses or grass-legume mixture. Whatever run-off comes from the cropped soil is retained by the strip of covercrop and this reduces soil erosion. Nitrogen fixing legumes also help in restoring soil fertility.

Fallow land – Land left without cropping to gain nutrients

- (vi) **Alley cropping:** It is a form of inter-cropping in which crops are planted between rows of trees or shrubs. This is also called Agro forestry.

Water Logging

- In order to provide congenial moisture to the growing crops, farmers usually apply heavy irrigation to their farmland. Also, in order to leach down the salts deeper into the soil, the farmer provides more irrigation.

LANDSLIDES

- Various anthropogenic activities like hydroelectric projects, large dams, reservoirs, construction of roads and railway lines, construction of buildings, mining etc are responsible for clearing of large forested areas.
- These result in landslide.

They also increase the turbidity of various nearby streams, thereby reducing their productivity.

About 29.32 % land degradation and desertification in India - 2021

- **DESERTIFICATION**
- Desertification is a process whereby the productive potential of arid or semiarid (receiving less rains) lands falls by ten percent or so.
- Severe desertification results in more than 50% drop in productivity and usually creates huge gullies and sand dunes.
- Desertification leads to the conversion of rangelands i.e., grasslands or irrigated croplands to desert like conditions in which agricultural productivity falls.

Desertification

- Desertification is characterized by **devegetation and loss of vegetal cover, depletion of groundwater, salinization and severe soil erosion.**

Causes of Desertification:

Formation of deserts may take place

(1). Due to natural phenomena like climate change or may be due to abusive use of land. (2). Anthropogenic activities.

The major anthropogenic activities responsible for desertification

- (a) Deforestation:
- (b) Overgrazing
- (c) Mining and quarrying.
- **Xenobiotics** are chemical substances that are foreign to animal life and thus includes such examples as plant constituents, drugs, pesticides, cosmetics, flavorings, fragrances, food additives, industrial chemicals and environmental pollutants.

THERE IS A NEED OF INDIVIDUAL EFFORTS IN THE CONSERVATION OF NATURAL RESOURCES.

Map of Desertification in India -



Conserve Water

Do not keep water taps running while brushing, shaving, washing or bath in the fountains.

In washing machines fill the machine only to the level required for your clothes.

I Install water-saving toilets that use not more than 6 liters per flush.

I Check for water leaks in pipes and toilets and repair them promptly. A small pin-hole sized leak will lead to the wastage of 640 liters of water in a month.

- I Reuse the soapy water of washings from clothes for washing off the courtyards, driveways etc.
- I Water the plants in your kitchen-garden and the lawns in the evening when evaporation losses are minimum. Never water
- the plants in mid-day.
- I Use drip irrigation and

and sprinkling irrigation to improve irrigation efficiency and reduce evaporation.

- I Install a small system to capture rain water and collect normally wasted used water from sinks, cloth-washers, bathtubs etc. which can be used for watering the plants.
- I Build rain water harvesting system in your house. Even the President of India is doing this.

ENERGY RESOURCES

- **Energy consumption of a nation is usually considered as an index of its development.**
- **This is because almost all the developmental activities are directly or indirectly dependent upon energy.**
- **Early man used it for cooking and heating purposes. Wind and hydropower have also been in use for the last 10,000 years.**
- **Germany – Falling water**

- The invention of steam engines replaced the burning of **wood** by coal and coal was later replaced to a great extent by **oil**. **In 1970 due** to Yom –Kippur War and Iranian revolution and Oil Politics – there was Oil Crisis.
- Due to Arab oil embargo the prices of oil shot up. This ultimately led to exploration and use of several alternate and nonconventional sources of energy - CES

GROWING ENERGY NEEDS

- Agriculture, industry, mining, transportation, lighting, cooling and heating in buildings all need energy.
- With INCREASE IN population the world is facing further energy deficit. The fossil fuels like coal, oil and natural gas which at present are supplying ALMOST 80 - 90% of the commercial energy of the world resources are not going to last for many more years.
- Our life style is changing very fast and from a simple way of life we are shifting to a luxurious life style.

- **Developed countries like U.S.A. and Canada constitute about 5% of the world's population but consume one fourth of global energy resources.**
- **An average person there consumes 300 GJ (Giga Joules, equal to 60 barrels of oils) per year. By contrast, an average man in a poor country like Bhutan, Nepal or Ethiopia consumes less than 1 GJ in a year.**

- **Global energy demand rose by 2.2% in 2024, reaching nearly 650 EJ.**
- **In 2023, India's primary energy consumption, listed as the third largest in the World, was 39.02 exajoules (EJ) behind China (170.74 EJ) and the USA (94.28 EJ). Per person, energy consumption in India is 27.3 gigajoules (GJ) compared to 120 GJ in China and 277.3 GJ in US in 2023.**

Exajoule is equal to 10^{18} joules

90 % villages in India electrified

- Today, India is the third largest energy consumer in the world.
- India has large population and Rural population (% of total population) in India was reported at 64.61 % in 2021 - according to the World Bank collection of development indicators, compiled from officially recognized sources.
- As of June 2025, India's total installed power capacity has reached a significant milestone with 476 GW, led by 240 GW of thermal, 110.9 GW of solar, and 51.3 GW of wind power, marking a strong shift towards renewable energy and energy security.

RENEWABLE AND NON-RENEWABLE ENERGY SOURCES

- **Renewable Resources which can be generated continuously in nature and are inexhaustible e.g. wood, solar energy, wind energy, tidal energy, hydropower, biomass energy, biofuels, geothermal energy and hydrogen.**
- **They are also known as non-conventional sources of energy and they can be used again and again in an endless manner**

Non-renewable Energy Resources

- **Non-renewable Resources which have accumulated in nature over a long span of time and cannot be quickly replenished when exhausted e.g. coal, petroleum, natural gas and nuclear fuels like uranium and thorium.**

It took millions of years for the coal to be formed out of wood and petroleum and natural gas to be formed from the dead marine animals.

(a) Renewable Energy Resource

- **Solar energy: Sun is the ultimate source of energy, directly or indirectly for all other forms of energy.**
- **Our Sun is about halfway through its life. But don't worry. It still has about 5,000,000,000—five billion—years to go.**
- **The solar energy received by the near earth space is approximately 1.4 kilojoules/second/m² known as solar constant.**

Solarimeter - is a pyranometer, a type of measuring device used to measure combined direct and diffuse solar radiation.

- **Traditionally, we have been using solar energy for drying clothes and food-grains, preservation of eatables and for obtaining salt from sea-water**
- **Solar heat collectors :**
- **These can be passive or active in nature. Passive solar heat collectors are natural materials like stones, bricks etc. or material like glass which absorb heat during the day time and release it slowly at night. Glaubers salt - Sodium sulphate decahydrate (32 deg. C).**

- **(iii) Solar cooker:**
- **Solar cookers make use of solar heat by reflecting the solar radiations using a mirror directly on to a glass sheet which covers the black insulated box within which the raw food is kept .**
- **(iv) Solar water heater:**
- **It consists of an insulated box painted black from inside and having a glass lid to receive and store solar heat. Inside the box it has black painted copper coil through which cold water is made to flow in, which gets heated and flows out into a storage tank.**
- **The hot water from the storage tank fitted on roof top is then supplied through pipes into buildings like hotels and hospitals.**

Active solar collectors pump a heat absorbing medium (air or water) through a small collector which is normally placed on the top of the building.

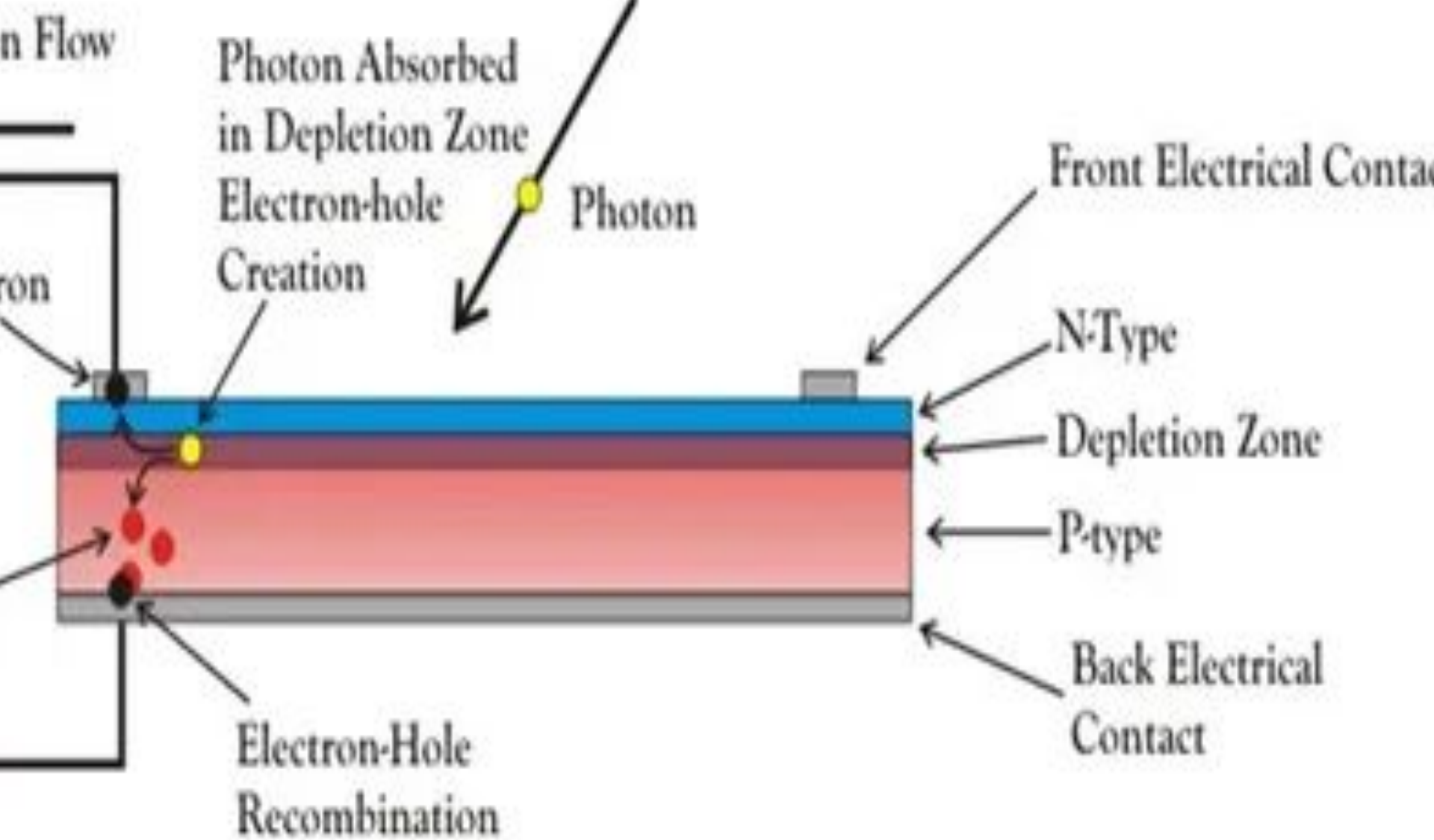
Solar cells/ PV cells :

They are also known as photovoltaic cells or PV cells. Solar cells are made of thin wafers of semi conductor materials like silicon. When solar radiations fall on them, a potential difference is produced which causes flow of electrons and produce electricity. – p-n junction.

Doping of solar cells

- Usually silicon is used in two layers, one layer being doped with boron, the other phosphorus. These layers have different chemical electric charges and subsequently both drive and direct the current of electrons.
- Phosphorus is n –type dopant, others – As, Sb
- (For generating electrons) .
- Boron is a p-type dopant, other is Indium
- (For moving positive holes to the n – type and receiving electrons from the p- type doped silicon semiconductor).

- **N-type materials increase the conductivity of a semiconductor by increasing the number of available electrons; P-type materials increase conductivity by increasing the number of holes present.**
- **P-N junctions are formed by joining n-type and p-type semiconductor materials, as shown in the figure in the next slide. Since the n-type region has a high electron concentration and the p-type a high hole concentration, electrons diffuse from the n-type side to the p-type side.**



Top Uses for Solar Energy

from  the spruce

Power Fans



Charge Batteries



Heat Home & Water



Heat Swimming Pool



Cook



Video on photovoltaic power generation

- https://www.google.com/search?q=video+on+photovo%3Btaic+power+generation+&sca_esv=5b71cfa89013eb9f&rlz=1C1YTUH_enIN1045IN1045&sxsrf=AE3TifP6eQMTH8lrMnA5swvxl_DoxK1Obw%3A1757958167557&ei=F1DlaKDjIZeqseMPls_02Qo&ved=0ahUKEwjgyu3AqNuPAxUXVWwGHZYnPasQ4dUDCBA&uact=5&oq=video+on+photovo%3Btaic+power+generation+&gs_lp=Egxnd3Mtd2l6LXNlcniAiJ3ZpZGVvIG9uIHBob3Rvdm87dGFpYyBwb3dlciBnZW5lcmF0aW9uIDIHECEYoAEYCjIHECEYoAEYCjIFECEYnwUyBRAhGJ8FSPuRAVAAWOOMAXABeAGQAQCYAbQBoAHmLaoBBDAuMzm4AQPIAQD4AQGYAiigArcvqAIKwglHECMYJxjqAsICChAjGIAEGCcYigXCAGQQIxgnwglLEAAAYgAQYkQIYigXCAGsQABiABBixAxiDAcICCBAAGIAEGLEDwglIEC4YgAQYsQPCAggQLhiABBjIBMICChAAGIAEGEMYigXCAGsQLhiABBixAxjIBMICDRAAGIAEGLEDGEMYigXCAGUQABiABMICChAAGIAEGAIYywhCAgcQABiABBgNwglGEAAAYDRgewglIEAAAYCBgNGB7CAgsQABiABBIGAXiKBcICCBAAGIAEGKIEwglFEAAAY7wWYAw3xBa-Si7ulAXoikgcEMS4zOaAHxIACsgcEMC4zObgHqS_CBwkWljE5LjE4LjPIB60B&sclient=gws-wiz-serp#fpstate=ive&vld=cid:28bd7006,vld:FQXkMt0Zm1A,st:0

Use of Fresnel lenses also - Cost ?

- The use of Fresnel lens is to magnify the light intensity from the sun to achieve higher solar collectability of solar panel which may increase power output.

(v) **Solar furnace:**

Here thousands of small plane mirrors are arranged in concave reflectors, all of which collect the solar heat and produce as high a temperature as 3000°C.

One may use concentrating lenses also but depends upon the economy.

Use of solar energy for solvent extractors and chemical reactions at IIT Delhi.

- **(vi) Solar power plant :**
- **Solar energy is harnessed on a large scale by using concave reflectors/ solar concentrators which cause boiling of water to produce steam. The steam turbine drives a generator to produce electricity. A solar power plant (50 K Watt capacity) has been installed at Gurugram, Haryana.**

Several solar photovoltaic systems are being installed in India lately at Karnataka, Telengana, Rajathan, Andhra Pradesh and Gujarat (Bhuj) etc.

- The Bhadla Solar Park is a solar power plant located in the Thar Desert of Rajasthan, India. It covers an area of 56 square kilometers and has a total installed capacity of 2,245 megawatts (MW), making it India's largest and the 11th-largest solar park in the world as of 2024.
- The NTPC REL. Khavda Solar Park (NTPC Renewable Energy Ltd.) is set to surpass the Bhadla Solar Park as the biggest solar power plant in India, with a proposed capacity of 4,750 MW. The construction of this facility is currently underway in Bhuj, Gujarat.

As of July 2025, India's cumulative solar power capacity stood at 119.02 GW, which includes 90.09 GW from ground-mounted solar plants, 19.88 GW from grid-connected rooftop systems, 3.06 GW from hybrid projects, and 5.09 GW from off-grid solar installations, reflecting the country's diverse approach to expanding



WIND ENERGY

- **The high speed winds have a lot of energy in them as kinetic energy due to their motion. The driving force of the winds is the sun. The wind energy is harnessed by making use of wind mills. The blades of the wind mill keep on rotating continuously due to the force of the striking wind.**

Height –10 – 200 meters

- These wind farms are ideally located in coastal regions, open grasslands or hilly regions, particularly mountain passes and ridges where the winds are strong and steady. The minimum wind speed required for satisfactory working of a wind generator is 15 km/hr.
- Global status: India has fourth largest wind power installed capacity and is third largest renewable energy producer in the world. Significant growth: From 21.04 GW in 2014 to 51.5 in May 2025. Role in Renewable Energy: Wind power is the second-largest contributor to India's renewable energy mix following solar pow

Wind Mills in India



HYDROPOWER

- The water flowing in a river is collected by constructing a big dam where the water is stored and allowed to fall from a height.
- The blades of the turbine located at the bottom of the dam move with the fast moving water which in turn rotate the generator and produces electricity. – Faradays law of Induction.
- The hydropower potential of India is estimated to be about 4×10 raised to the power 11KW-hours.
- Till now we have utilized only a little more than 11% of this potential.

Hydroelectric Power Generation



TIDAL ENERGY

- Ocean tides produced by gravitational forces of sun and moon contain enormous amounts of energy. The high tide and low tide refer to the rise and fall of water in the oceans.
- There are only a few sites in the world where tidal energy can be suitably harnessed. The bay of Fundy Canada having 17-18 m high tides has a potential of 5,000 MW of power generation.
- **Wave energy** can also be used for power generation.

- **The tidal mill at La Rance, France is one of the first modern tidal power mill.**
- **In India Gulf of Cambay, Gulf of Kutch and the Sunder bans deltas are the tidal power sites.**

OCEAN THERMAL ENERGY CONVERSION (OTEC)

- **The energy available due to the difference in temperature of water at the surface of the tropical oceans and at deeper levels is called Ocean Thermal Energy. A difference of 20°C or more is required between surface water and deeper water of ocean for operating OTEC (Ocean Thermal Energy Conversion) power plants.**
- **The warm surface water of ocean is used to boil a liquid like ammonia.**

- **The high pressure vapours of the liquid formed by boiling liquid are then used to turn the turbine of a generator and produce electricity – Rankine cycle.**
- **The colder water from the deeper oceans is pumped to cool and condense the vapours into liquid. Thus the process keeps on going continuously for 24 hours a day.**

The ten countries with the most renewable energy are Iceland, Norway, Sweden, Brazil, New Zealand, Denmark, Austria, Switzerland, Finland, and Colombia

- **Renewables 2022 Global Status Report (GSR) : India Ranks **Third** Globally, After China And USA, For Total Renewable Power Addition In 2021.**
- **As per IRENA RE Statistics 2025, India ranks 4th globally in Renewable Energy Installed Capacity, 4th in Wind Power, and 3rd in Solar Power capacity. Expansion in Renewable Energy Capacity: Installed RE capacity has grown from 76.37 GW in March 2014 to 226.79 GW in June 2025, an increase of nearly 3 times.**

GEOHERMAL ENERGY

- The energy harnessed from the hot rocks present inside the earth is called **geothermal energy**. High temperature, high pressure steam fields exist below the earth's surface in many places.
- In some places, **the steam or the hot water comes out of the ground naturally through cracks** in the form of natural **geysers** as in **Manikaran, Kullu and Sohna, Haryana**.

Jharia ?

- In USA and New Zealand, there are several geothermal plants working successfully.

BIOMASS ENERGY

(a) Energy Plantations:

Fast growing trees like cottonwood, poplar and Leucaena, non-woody herbaceous grasses, crop plants like sugarcane, sweet sorghum and sugar beet, aquatic weeds like water hyacinth

and sea-weeds and carbohydrate rich potato, cereal etc. are some of the important energy plantations.

- **(b) Agricultural and Urban Waste biomass:**
Crop residues, bagasse (sugarcane residues), coconut shells, peanut hulls, cotton stalks
etc. are some of the common agricultural wastes which produce energy by burning.

- **Animal dung, fishery and poultry waste and even human refuse are examples of biomass energy.**
- **In Brazil 30 % of electricity is obtained from burning bagasse.**
- **In rural India, animal dung cakes are burnt to produce heat. About 80 % of rural heat energy requirements are/ were met by burning agricultural wastes, wood and animal dung cakes – CBG and LNG Cylinders.**

Compressed BIOGAS Cylinders

- Biogas is a mixture of methane, carbon dioxide, hydrogen and hydrogen sulphide, the major constituent being methane. Biogas is produced by anaerobic digestion of animal wastes (some times plant wastes) in the presence of water.
- Biogas plants used in our country are basically of two types:
- 1. Floating gas-holder type and 2. Fixed Dome type- Compressed Biogas – Removal of CO₂.

Biogas Plant in India



BIOFUELS

- **Ethanol can be easily produced from carbohydrate rich substances like sugarcane. It burns clean and is non-polluting.**
- **However, as compared to petrol its calorific value is less and therefore, produces much less heat than petrol.**
- **Methanol is very useful since it burns at a lower temperature than gasoline or diesel.**

One hundred years ago, Rudolf Diesel tested peanut oil as fuel for his engine for the first time on August 10, 1893. In the 1930s and 1940s vegetable oils were used as diesel fuels from time to time, usually only in emergency.

- Biodiesel and Green Diesel :
- From seed oils : Sunflower oil, Soyabean oil, Castor oil, Jatropha oil, Pongamia oil etc.
- Waste cooking oil.
- HYDROGEN AS A FUEL
- Electrolysis of water using power generated by renewable energy.
- Green Hydrogen.
- Blue Hydrogen
- Brown Hydrogen
- Pink Hydrogen.

b) Non-Renewable Energy Sources

- **COAL**
- Coal was formed 255-350 million years ago in the hot, damp regions of the earth during the carboniferous age.
- At the present rate of usage, the coal reserves are likely to last for about 200 years and if its use increases by 2% per year, then it will last for another 65 years?

- **Peat**
- **Lignite/ Brown coal. Bituminous coal**
- **Anthracite**
- **Production of coal during 2022-23 were 800 MT (Provisional) with a positive growth of 8.67%.**

- **India has almost 9% of the world's total coal reserves and it has 360 Billion tons).**
- **Major coal fields in India are Raniganj, Jharia, Bokaro, Singrauli, and Godavari valley.**

Coking coal is mainly imported in India. About 58 million tons of coking coal is imported by India and almost same amount was produced by India.

Coal Capital - Dhanbad

- **The coal states of India are Jharkhand, Orissa, West Bengal, Madhya Pradesh, Andhra Pradesh and Maharashtra. Anthracite coal occurs only in J & K.**
- **When coal is burnt it produces carbon dioxide, which is a greenhouse gas responsible for causing enhanced global warming. Fly ash is also produced.**
- **There are clean coal technologies.**

- Coal also contains impurities like sulphur and therefore as it burns the smoke contains toxic gases like oxides of sulphur and nitrogen.
- **PETROLEUM**
- It is the lifeline of global economy. There are 13 countries in the world having 67% of the petroleum reserves which together form the OPEC (Organization of Petroleum exporting countries). About $1/4^{\text{th}}$ of the oil reserves are in Saudi Arabia.

Petroleum Refining

- At the present rate of usage, the world's crude oil reserves are estimated to get exhausted in just 40 years. Some optimists, however, believe that there are some yet undiscovered reserves. Even then the crude oil reserves will last for another 40-75 years or so.

Crude petroleum

- Crude oil is a complex mixture of alkane and aromatic hydrocarbons. Hence it has to be purified and refined by the process of fractional distillation, during which process different constituents. – Gasoline, Diesel, LPG ATF etc. are produced.

Refining – Removing S and other impurities from crude oil

- Oil fractions separate out at different temperatures. (Petroleum refineries) We get a large variety of products from this, namely, petroleum LPGs, kerosene, petrol, diesel, fuel oil, lubricating oil, paraffin wax, asphalt, plastic etc - Petrochemicals
- Petroleum is a cleaner fuel as compared to coal as it burns completely and leaves no residue. It is also easier to transport and use. That is the reason why petroleum is preferred amongst all the fossil fuels – IOCL. HPC, BP etc.

Liquefied petroleum gas (LPG):

- The main component of petroleum is butane, the other being propane and ethane. The petroleum gas is easily converted to liquid form under pressure as LPG.

NATURAL GAS

It is mainly composed of methane (95%) with small amounts of propane and ethane.

Shale gas (70-90 % methane) from shale formations.

- It is a fossil fuel. Natural gas deposits mostly accompany oil deposits because it has been formed by decomposing remains of dead animals and plants buried under the earth.
- Natural gas is the cleanest fossil fuel. It can be easily transported through pipelines. It has a high calorific value of about 34 to 52 MJ/m³ and burns without any smoke.

LNG imported and used as CNG

- **Russia has maximum reserves (40 %), followed by Iran (14%) and USA (7%).**
- **Natural gas reserves are found in association with all the oil fields in India. Some new gas fields have been found in Tripura, Jaisalmer, Off-shore area of Mumbai and the Krishna Godavari Delta - ONGC and GAIL.**

NUCLEAR ENERGY

- Nuclear energy is known for its high destructive power as evidenced from nuclear weapons. The nuclear energy can also be harnessed for providing commercial energy.

Nuclear energy can be generated by two types of reactions:

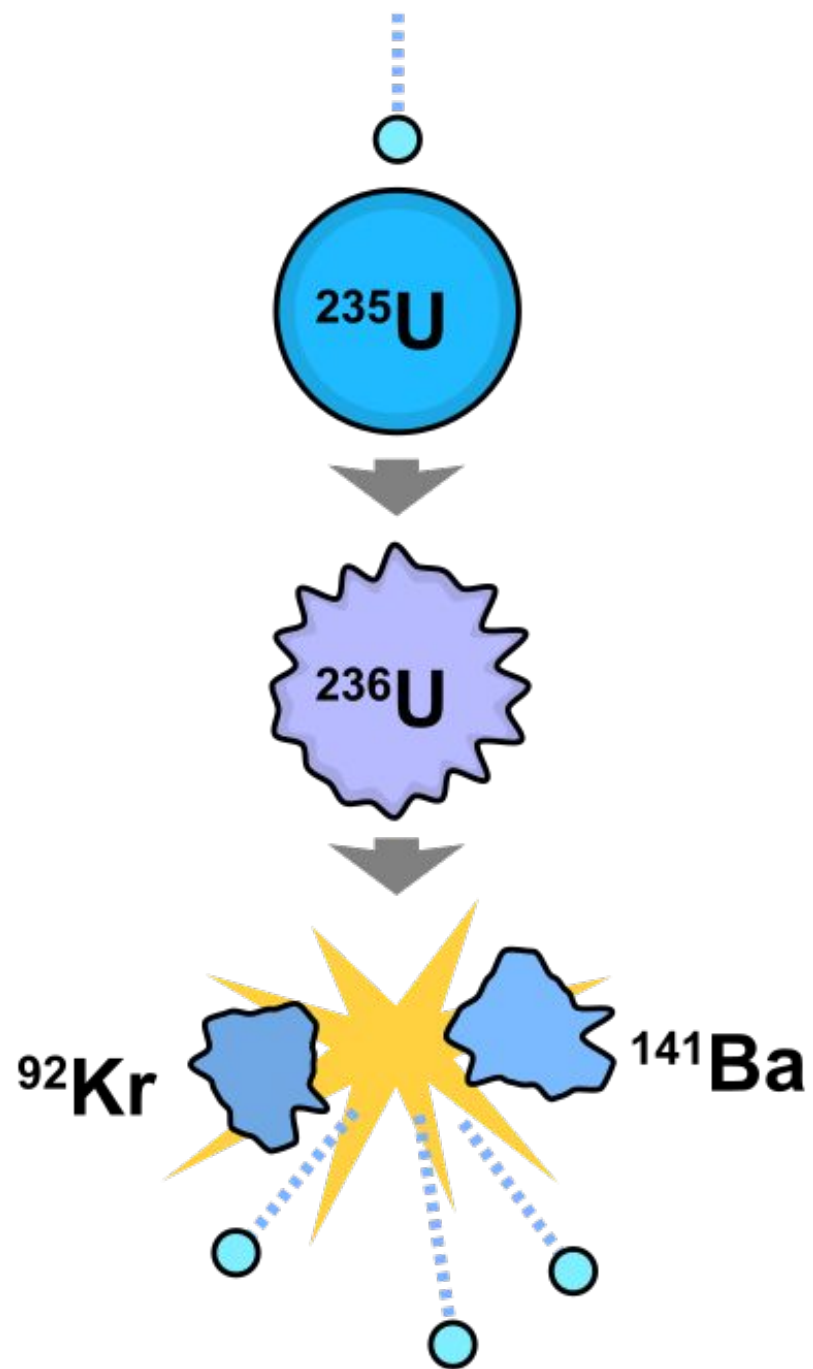
- **(i) Nuclear Fission:** It is the nuclear change in which nucleus of certain isotopes with large mass numbers are split into

Nuclear Fuels

lighter nuclei on bombardment by neutrons and a large amount of energy is released through a chain reaction

- ${}_{92}\text{U}^{235} + {}_0\text{n}^1 \rightarrow {}_{36}\text{Kr}^{92} + {}_{56}\text{Ba}^{141} + 3 {}_0\text{n}^1 + \text{Energy}$
- $\text{U } 238 + 0 \text{ Nu } 1 = \text{Pu } 239 + \text{Energy}$

- A typical nuclear power plant in operation today uses about 2 kg Uranium-235 to generate 1000 megawatts of electricity.
- About 5600 tons (5.1×10^6 kg) of coal are required to produce the same amount of electricity in a conventional power plant.
- Presently, India has 22 operating nuclear reactors, with an installed capacity of 6780 MWe.



(ii) Nuclear fusion:

- Here two isotopes of a light element are forced together at extremely high temperatures (1 billion °C) until they fuse to form a heavier nucleus releasing enormous energy in the process. It is difficult to initiate the process but it releases more energy than nuclear fission –
Chaos- **Containment of heat**
- There are around 20 fusion reactors in the world, all striving to reach the extremely high temperatures needed and R & D work is in progress.

- **Nuclear Fusion reactions power the Sun and other stars. In a fusion reaction, two light nuclei merge to form a single heavier nucleus. The process releases energy because the total mass of the resulting single nucleus is less than the mass of the two original nuclei. The leftover mass becomes energy.**

D

Neutron

Energy

Fusion

He

T



Present Energy Scenario

- Present trend is to go in for the cleaner renewable energy sources such as solar energy mainly.
- Use of Electrical vehicles is being encouraged which may be fueled by Fuel cells and Hydrogen..
- During COP 21 at Paris some voluntary commitments to reduce CO₂ emissions were made - Nationally Determined Contributions (NDCs) - Lifestyle for Environment' as a key to combating climate change“.

What is India's NDC target by 2030?

- India's NDC has three main elements:
- An emissions-intensity target of 45% below 2005 levels by 2030.
- A target of achieving 50% cumulative electric power installed capacity from non-fossil fuel-based energy resources by 2030.
- Hon'ble Prime Minister's vision of sustainable lifestyles and climate justice to protect the poor and vulnerable from adverse impacts of climate change

EQUITABLE USE OF RESOURCES FOR SUSTAINABLE LIFE

STYLE – Background Information

- **There is a big divide in the world as North and South, the more developed countries (MDC's) and less developed countries (LDC's), the haves and the have nots..**
- **The MDC's have only 22% of world's population, but they use 88% of its natural resources, 73% of its energy and command 85% of its income. In turn, they contribute a very big proportion to its pollution.**
- **These countries include USA, Canada, Japan, the CIS, Australia , New Zealand and Western European Countries.**

- **The LDCs on the other hand, have very low or moderate industrial growth, have 78% of the world population and use about 12% of natural resources and 27% of energy. Their income is merely 15% of global income. The gap between the two is increasing with time due to sharp increase in population in the LDC's. The rich have grown richer while the poor have stayed poor or gone even poorer**

- **As the rich nations are developing more, they are also leading to more pollution and sustainability of the earth's life support system is under threat.**
- **As the rich nations continue to grow, they will reach a limit. If they have a growth rate of 10 % every year, they will show 1024 times increase in the next 70 years or so.**
- **Will this much of growth be sustainable?**

- **Thus, the two basic causes of unsustainability are over population in poor countries who have under consumption of resources and over consumption of resources by the rich countries, which generate wastes.**
- **In order to achieve sustainable life styles it is desirable to achieve a more balanced and equitable distribution of global resources and income to meet everyone's basic needs.**

Top 10: Sustainable Countries

- Canada.
- Germany.
- Norway.
- Switzerland.
- Finland.
- Rural Sweden.
- Stockholm, Sweden.
- Denmark.